

ForgeCloud Observability, Security & Reliability

Metrics, logs, traces, controls, autoscaling and failure testing

1. Observability Architecture

```
Applications / Platform
|
+--> Metrics ----> Prometheus ----> Grafana
|
+--> Logs -----> CloudWatch / log view
|
+--> Traces ----> OpenTelemetry
|
+--> Operational status -----> ForgeCloud dashboard
```

2. Signals

Signal	Examples
Metrics	CPU, memory, request count, request rate, error rate, latency, P95 latency, pod count
Logs	Application events, errors, deployment events
Traces	Request path and service timing where distributed tracing is enabled

3. Security Architecture

- JWT authentication.
- Role-based access control.
- Password hashing.
- Least-privilege AWS IAM.
- Kubernetes RBAC.
- Secrets management.
- CORS and input validation.
- API authorization.
- Audit logging.
- AWS security groups.
- Kubernetes network policies where practical.

4. Reliability Experiments

Experiment	Expected behavior	Measurements
Terminate pod	Replacement pod starts and desired replica count recovers	Recovery time, availability, errors
Deploy unhealthy version	Health check fails and previous version can be restored	Rollback time, error rate, latency
Increase traffic	HPA increases replicas	CPU, replicas, P95 latency, throughput

5. Evidence-Based Metrics

- Deployment time.
- API latency.
- P95 application latency.
- Throughput under load.
- Autoscaling response time.
- Resource utilization.
- Rollback/recovery time.
- Error rate.
- Application availability.
- Infrastructure provisioning success rate.